Open-access practical guide for the fabrication of cardiac anatomical models using additive manufacturing

1. Introduction and Purpose of the Guide

This guide describes a reproducible workflow for the acquisition, preparation, manufacturing, post-processing, and validation of three-dimensionally printed anatomical cardiac biomodels intended primarily for educational and anatomical simulation applications.

The development of cardiac biomodels using additive manufacturing has made it possible to tangibly represent complex cardiac structures, facilitating the spatial understanding of structural heart diseases that are often difficult to interpret using only two-dimensional medical images. However, one of the main obstacles to their adoption in academic settings remains the lack of clear and reproducible guidelines documenting the entire manufacturing process.

In response to this need, this appendix proposes a reproducible, open-access guide that documents the workflow for manufacturing cardiac biomodels, including the following stages:

- Obtaining the digital model
- Preparing the model for manufacturing
- Preparing the file for printing
- Fabrication using additive manufacturing
- Post-processing of the biomodel
- Anatomical and functional evaluation

This guide is based on the experience gained during a final graduation project focused on the development of physical models of hearts with structural heart diseases using additive manufacturing and flexible polymeric materials.

Although the research used a pediatric heart with Tetralogy of Fallot as a case study, the procedure described can be applied to any cardiac model available in a digital format suitable for 3D printing.

The guide specifies Formlabs Silicone 40A elastomer as the manufacturing material, selected during the project's experimental phase for its mechanical properties suitable for simulating soft tissues. It is important to note that, at the time of this guide's preparation, this material is commercially available only in black, which may limit the visual differentiation of certain anatomical structures. This characteristic is common in elastomeric materials recently introduced to the market and could be modified in future versions of the material through

pigmented variants. Alternatives such as the use of coatings or surface pigmentation of the model are not within the scope of this work.

2. Scope of the Guide

This guide is intended for the fabrication of flexible cardiac biomodels for non-clinical purposes using additive manufacturing, digital anatomical models, and Formlabs' Silicone 40A elastomeric material.

Its intended applications include:

- Anatomical education
- Demonstration of congenital heart conditions
- Basic hands-on training in handling cardiac structures
- Simulation of manipulation and suturing on simulated tissues
- Visual support in academic, teaching, or technical communication contexts

The guide can be applied to anatomical models representing structural heart diseases, provided that the digital file has sufficient geometric quality for three-dimensional preparation and printing.

The following are not within the scope of this document:

- The use of the biomodel for personalized surgical planning for specific patients
- Direct clinical decision-making based on the physical model

3. Rationale and justification for the proposed workflow

The proposed workflow is based on the results obtained during the project's experimental phase, in which various additive manufacturing technologies and flexible materials were evaluated for the fabrication of anatomical cardiac models.

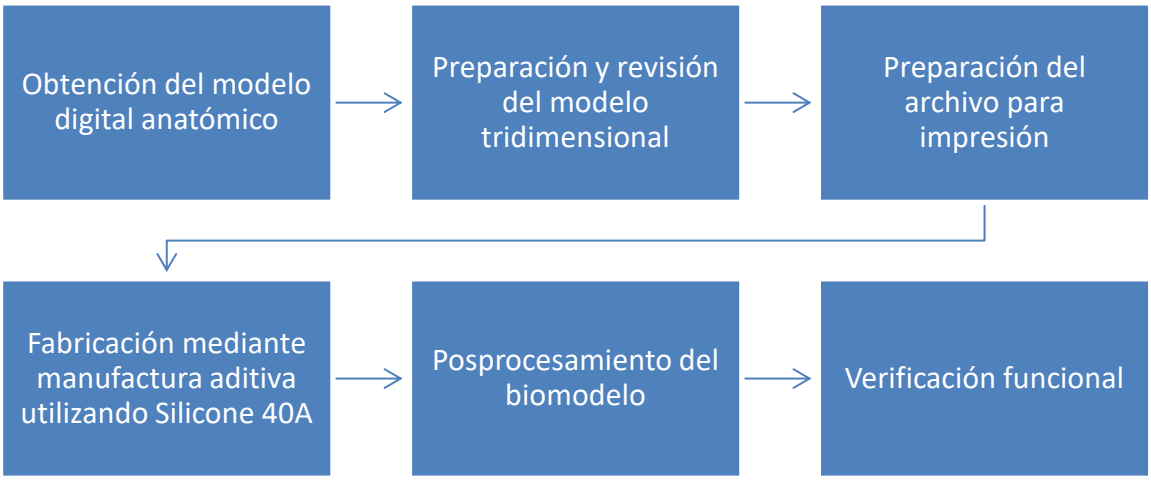
Technologies accessible in academic settings were analyzed, such as FDM with flexible materials (e.g., TPU) and elastomeric photopolymerizable resin systems. Although these alternatives offer advantages in terms of cost and availability, they presented limitations in surface fidelity, precision in complex geometries, and behavior during handling.

PolyJet technology, recognized for its high resolution and surface quality, was also evaluated. However, its use proved less practical due to the limited availability of equipment, high operating costs, and reliance on specialized suppliers. Furthermore, in the proof-of-concept tests conducted during the project, the materials evaluated using this technology showed

limited performance in cutting and suturing tests, which affects their usefulness in handling and training activities.

Based on this comparative evaluation, Formlabs' Silicone 40A material was selected, as it demonstrated an appropriate balance between geometric fidelity, elastomeric behavior, and resistance to repeated manipulation. Based on these results, the guide proposes a simplified workflow that allows for the reproduction of the biomodel's fabrication without the need to repeat the entire material and technology comparison phase.

Consequently, the workflow is structured into the following main stages:



4. Components of the simulation system

The physical system described in this guide consists of three main elements that allow the cardiac biomodel to be integrated into a stable manipulation environment:

- Flexible cardiac biomodel
- Anatomical support base
- Structural thoracic cover (optional depending on application)

Each component fulfills a specific function within the simulation system.

The flexible cardiac biomodel is the main anatomical component and is manufactured using additive manufacturing with Silicone 40A.

The anatomical support base allows the heart to be positioned stably during manipulation, demonstration, or training activities.

The structural thoracic cover can be used to partially represent the thoracic cavity, providing additional anatomical context during educational demonstrations.

5. Obtaining digital anatomical models

5.1. Source of the models

The digital models used in this guide come from repositories specializing in cardiac anatomy that are used for research and medical education.

This study uses models from the 3D Heart Project repository, which are already segmented and prepared for educational and research use.

The use of pre-segmented models allows the work to focus on the manufacturing and validation process of the physical biomodel, significantly reducing the time associated with medical image processing.

5.2. Model Format

Digital models must be in one of the following formats compatible with additive manufacturing:

- STL
- OBJ
- PLY

The STL format is the most commonly used format in additive manufacturing and will be the primary format used in this guide.

6. Preparing the Model for Manufacturing

Preparing the digital model is a critical step to ensure that the anatomical geometry can be correctly fabricated using additive manufacturing and that the final biomodel exhibits sufficient mechanical stability during handling. This section describes the steps used to prepare the cardiac model using Meshmixer, a widely used software for editing and repairing three-dimensional meshes.

The objective of this stage is to ensure that the model meets the following requirements prior to fabrication:

- Geometric integrity of the mesh (watertight model)

- Absence of topological errors
- Structural stability of the printed model
- Preservation of relevant anatomical structures

The steps described below correspond to the workflow used in the preparation of cardiac biomodels.

6.1. Mesh inspection and repair (Watertight)

Before preparing the model for printing, it is necessary to verify that the geometry is completely closed (watertight). An open model can cause failures during the printing process or produce incorrect geometric results.

Procedure in Meshmixer

1. Import the model file in STL format into Meshmixer.
2. Go to the menu:

   Analysis → Inspector

3. The system will automatically perform a mesh analysis and highlight potential errors using color indicators.

   The most common errors include:

   - Holes in the surface
   - Open edges
   - Geometry intersections
   - Inverted faces
4. Select **Auto Repair All** to automatically close small openings in the mesh.

In the case of larger openings or complex defects, manual corrections may be necessary using surface editing tools.

Once the repair is complete, the model should be a fully closed surface, a necessary condition for correct volume generation during printing.

6.2. Selective Surface Smoothing

Models derived from medical segmentation processes or digital repositories may exhibit surface irregularities or geometric artifacts. Surface smoothing allows for improved model quality without altering its overall anatomical shape.

However, it is essential to avoid modifications to critical anatomical structures, especially in regions such as the interventricular septum, where small geometric alterations could affect the anatomical representation of the pathology.

Procedure in Meshmixer

1. Access the menu:

Select

2. Use the selection tools to identify surface areas with irregularities.
3. Go to:

Sculpt → Smooth

4. Apply smoothing locally and progressively, using moderate intensity values.

During this process, it is recommended that you:

- Apply smoothing only to external surfaces
- Avoid areas of anatomical interest
- Do not modify the geometry of the interventricular septum

The goal is to improve surface continuity without compromising the anatomical accuracy of the model.

6.3. Model scaling

Depending on the source of the digital model, it may be necessary to verify or adjust the scale to ensure that the biomodel accurately represents the anatomical dimensions.

Procedure in Meshmixer

1. Go to:

Edit → Transform

2. Review the model's current dimensions.
3. If necessary, apply a uniform scale factor to adjust the model's size.

It is recommended to maintain uniform scaling along all three axes (X, Y, Z) to avoid geometric distortions.

In educational applications, the model can be kept at actual scale or adapted to dimensions that facilitate its manipulation during demonstrations.

6.4. Generating minimum wall thickness

Models derived from medical segmentation may contain extremely thin surfaces that are structurally unstable during printing with elastomeric materials.

To ensure the stability of the biomodel manufactured with Silicone 40A, a minimum wall thickness of 2 mm is established.

This thickness allows for:

- Maintain the structural integrity of the model
- Prevent deformation during handling
- Improve resistance to shear or suture forces

Procedure in Meshmixer

1. Access the menu:

Edit → Make Solid

2. Set the main parameters:

Solid Type: Accurate

Solid Accuracy: high

Mesh Density: medium or high

Minimum Thickness: 2 mm

3. Apply the operation to generate a solid geometry with uniform thickness.

It is recommended to review the geometry again using:

Analysis → Inspector

to confirm that the resulting model remains completely closed and free of topological errors.

6.5. Exporting the prepared model

Once the repair, smoothing, scaling, and thickness generation steps are complete, the model can be exported again in STL format for preparation in the corresponding printing software.

This file constitutes the final version of the digital model, ready for the print preparation and additive manufacturing stage.

7. Manufacturing of the flexible cardiac biomodel

The flexible cardiac biomodel is the main anatomical component of the simulation system. It is manufactured using additive manufacturing with **Formlabs' Silicone 40A** , an elastomer

specifically designed for applications requiring flexibility, tear resistance, and geometric stability after curing.

The model is manufactured using resin printers from the Formlabs Form 4 platform, with **PreForm** software used to prepare the print file. At this stage, it is essential to properly configure the model's orientation, support generation, and print parameters to ensure the geometric quality of the biomodel and avoid defects associated with the behavior of elastomeric materials during the manufacturing process.

7.1. File Preparation in PreForm

The model, previously prepared in STL format, must be imported into the PreForm software, which is used to configure the printing process on Formlabs systems.

General procedure

1. Open PreForm.
2. Import the STL file using the option:

File → Open

3. Select the printing material:

Material → Silicone 40A

When you select the material, the software automatically adjusts the print profiles compatible with this elastomer, including the exposure strategy and layer settings used by the printer.

Next, proceed with model orientation and support generation, aspects that have a direct impact on the final quality of the part.

7.2. Model Orientation

Model orientation within the print volume is a critical factor when working with elastomeric materials, as improper orientation can lead to defects such as:

- Trapped liquid resin
- Internal bubble formation
- Deformations during curing
- Collapse of thin surfaces

To minimize these risks, it is recommended to orient the model so that internal cavities facilitate resin drainage during printing, preventing areas where liquid material could become trapped. Additionally, it is advisable to avoid having large surfaces that are completely parallel to the print bed, as this can generate additional stresses during the process. Finally, care should be taken to ensure that the most delicate anatomical regions of the model have adequate support to guarantee their stability and preserve the accuracy of the geometry during manufacturing.

In the case of cardiac models, it is usually advisable to apply a moderate tilt to the model (approximately between 20° and 40°) relative to the printing platform. This tilt facilitates:

- Resin drainage from internal cavities
- A more uniform distribution of release forces
- A reduction in the risk of bubble formation or local collapses

The final orientation should be visually evaluated in PreForm to ensure that complex anatomical structures (e.g., vessels or cavities) are not oriented in a way that could trap liquid resin during the process.

7.3. Support Generation for Elastomeric Materials

Elastomeric materials behave differently from rigid resins during printing. Due to their flexibility, parts can deform more easily during the release phases of each layer.

For this reason, it is recommended to use a higher support density than is typically used for rigid materials.

Recommended settings

In PreForm, supports can be generated using the option:

Supports → Auto Generate Selected

It is recommended to manually review the generated supports afterward.

General recommendations include:

- Use **medium to high** support density
- Reinforce the lower areas of the model
- Add additional supports on large surfaces
- Ensure support for thin or overhanging geometries

It is also important to avoid, as much as possible, placing supports in critical anatomical areas that must remain visually clean to facilitate subsequent observation or manipulation. Proper support placement helps reduce warping during the printing process, maintain model stability throughout manufacturing, and improve the geometric accuracy of the final biomodel.

7.4. Print settings in Form 4 for Silicone 40A

The print parameters used for Silicone 40A on the Form 4 printer are preconfigured within the material profile in PreForm. However, it is important to understand the key parameters that influence the manufacturing process.

The most relevant parameters include:

Parameter	Recommended settings
Printing technology	LFD (Low Force Display)
Layer height	Automatically defined by the material profile
Material	Silicone 40A
Exposure strategy	Automatically controlled by PreForm
Supports	Automatically generated and manually adjusted
Post-curing	Described in Section 9 of this guide

The LFD (Low Force Display) technology used by the Form 4 reduces detachment forces during printing, which is particularly important when manufacturing flexible parts or complex geometries.

Once these parameters have been configured and the model orientation verified, the file can be sent to the printer to start the manufacturing process for the flexible cardiac biomodel.

It is important that the model has been previously prepared with a minimum wall thickness of 2 mm, as described in Section 6.4. Thinner walls may cause deformation during printing or compromise the integrity of the biomodel during handling.

Correct configuration of these variables allows for the production of biomodels with high anatomical fidelity, structural stability, and mechanical behavior suitable for simulation and manipulation activities.

8. Fabrication of the Anatomical Support Base and Structural Components

The structural components of the system have been designed to provide mechanical stability to the flexible biomodel and contextualize the simulation practice. These files are available in the attached repository in STL format and are optimized for fabrication using Fused Deposition Modeling (FDM).

8.1. Recommended material

PLA is the standard material for manufacturing the support base and thoracic cover.

PLA offers high structural rigidity, low thermal shrinkage (warping), and ease of printing on conventional equipment. Since these parts do not require contact with biological fluids or aggressive sterilization processes, PLA ensures the system's stability at minimal cost.

8.2. Print Settings (Slicing)

To ensure that the fastening mechanisms (specifically the "claw" system and pivot points) maintain their integrity during user handling, the following print parameters are recommended:

Parameter	Recommended Setting
Layer height	0.2 mm (Balance between resolution and time)
Wall thickness (Perimeters)	Minimum 3 layers (Essential for joint strength)
Infill	15%–20%
Infill pattern	Helical or Grid (Provides multidirectional strength)
Top/bottom layers	4 layers
Nozzle temperature	As per manufacturer's specifications (Typically 200°C - 215°C)

8.3. Clamping design considerations

The anatomical support base features a claw-type clamping mechanism specifically designed to interact with low-hardness elastomers (such as 40A silicone).

Mechanical adjustment: After printing, verify that the mechanism's clamps move freely. In case of excessive friction due to printer tolerances, light sanding of the shaft contact areas is recommended.

Interface with the biomodel: The design allows the heart to be "embraced" by the claw, distributing contact pressure to prevent tears in the flexible material during suturing or manipulation.

8.4. Assembly with the pediatric thoracic cover

Once the structural components have been manufactured using FDM printing, the simulation system is assembled, integrating the anatomical support base, the flexible cardiac biomodel, and the pediatric thoracic cover.

The system's design allows for modular assembly, so that the user can use the heart with just the support base or additionally integrate the thoracic cage when a more complete anatomical context is required.

The recommended assembly procedure is as follows:

1. **Installing the biomodel on the support base**

The flexible cardiac biomodel should be placed between the claws of the clamping mechanism. The clamps of the claw-type system are designed to exert moderate pressure to stabilize the model without causing permanent deformation of the elastomer.

2. **Verification of stability**

Once the biomodel is positioned, verify that the model remains stable during manipulation. The system should allow for moderate manual movements without the heart shifting or rotating within the clamping mechanism.

3. **Assembly of the pediatric thoracic cover**

The thoracic cover can be installed by aligning the designed attachment points on the structural base. These attachment points allow the cover to be positioned consistently, maintaining the system's correct anatomical orientation.

4. **Final adjustment of the system**

After assembly, it is recommended to verify that:

- The cover does not interfere with access to the biomodel during practice
- The biomodel remains visible from the main observation angles

- The structural components do not exert excessive compression on the elastomeric material

This integration system allows for the recreation of a simplified anatomical environment of the pediatric thorax, facilitating spatial understanding of cardiac structures and enhancing the simulation experience during educational or demonstration activities.

The system's modular design also allows components to be easily disassembled for maintenance, replacement of the biomodel, or transport of the system, while maintaining the overall functionality of the assembly.

9. Post-processing of the biomodel

Post-processing is a critical step in the fabrication of biomodels printed with Silicone 40A, as it largely determines the final mechanical properties of the material, the surface quality of the model, and the removal of uncured material residue.

Due to the elastomeric nature of the material, the cleaning procedure differs from that of other photopolymers used in additive manufacturing. This section describes the recommendations based on Formlabs' technical guidelines, adapted to the workflow of this guide.

9.1. Washing the Printed Biomodel

According to the manufacturer's recommendations, the standard cleaning procedure for Silicone 40A consists of washing using a solvent mixture within a Form Wash system.

The standard procedure recommended by Formlabs uses:

20% n-butyl acetate + 80% isopropyl alcohol (IPA) for 20 minutes.

However, due to the characteristic odor of n-butyl acetate, the manufacturer also describes an alternative procedure using only IPA and ultrasonic cleaning, which was considered in this workflow.

Alternative procedure with IPA and ultrasonic bath

1. Remove the model from the print bed.
2. Place the parts in a container with enough IPA to completely cover the model.
3. Place the container inside an ultrasonic bath filled with water.

Formlabs recommends using ultrasonic cleaners in the range of:

35–45 kHz

4. Sonicate the parts for 2 to 3 minutes.

During washing, it is normal to observe that the IPA becomes cloudy or that silicone residues appear in suspension, since silicone is not soluble in IPA.

5. Repeat the process using clean IPA in a second container to ensure complete removal of residues.

A properly cleaned part should not have any visible silicone residue on its surface.

6. After washing, allow the parts to air dry for at least 30 minutes before beginning the curing process.

9.2. Removal of Support Structures

Removing supports from elastomeric materials requires greater care than with rigid resins, as the material can deform or tear if excessive force is applied.

It is recommended to remove the supports after washing, when the material still retains some flexibility.

Recommended tools:

- Precision tweezers
- Modeling cutters
- Scalpel or fine blade

Handling recommendations:

- Remove the main supports first
- Avoid sudden pulling
- Cut the supports close to their point of contact

The flexibility of 40A silicone allows the part to be bent slightly to access supports located in internal areas.

9.3. Final curing of the biomodel

After washing and removing the supports, the model must undergo a thermal and UV curing process to achieve its final mechanical properties.

The manufacturer recommends using the Form Cure system.

Recommended settings by Formlabs

Parameter	Settings

Material	Silicone 40A
Temperature	60 °C
Time	45 minutes

Procedure:

1. Place the parts in a glass container or UV-transparent container.
2. Completely submerge the parts in water.
3. Place the container inside the Form Cure unit.
4. Run the 45-minute cycle at 60 °C.

Immersion in water during curing helps to:

- Improve heat transfer
- Promote uniform polymerization
- Reduce surface defects

The manufacturer states that additional curing does not adversely affect the material, especially in thick or bulky parts.

10. Integration and assembly of the simulation system

Once the manufacturing and post-processing stages of the various components are complete, the simulation system is integrated. This stage involves assembling the flexible cardiac biomodel made of Silicone 40A with the anatomical support base printed in PLA and, when required, incorporating the structural thoracic cover.

The objective of this phase is to ensure that the biomodel remains stable during handling, maintains its correct anatomical orientation, and allows for the recreation of a working environment that facilitates demonstration, training, or basic simulation activities.

The system has been designed with a modular approach, allowing each component to be easily disassembled for maintenance, transport, or replacement of the biomodel.

10.1. Assembling the biomodel onto the support base

The first step in assembly consists of securing the flexible cardiac biomodel within the claw-type clamping mechanism located on the anatomical support base.

This mechanism is designed to interact with low-hardness elastomeric materials, such as 40A silicone, allowing the model to be secured without compromising its structural integrity.

Assembly procedure:

1. Place the cardiac biomodel between the clamps of the claw-type clamping system.
2. Gradually adjust the opening of the clamps until the model is firmly held in place.
3. Verify that the pressure applied is sufficient to prevent the model from shifting, but without excessively compressing the elastomeric material.

Due to the flexible nature of Silicone 40A, excessive compression can cause deformations in the anatomical geometry and alterations in the shape of the internal cavities. For this reason, the clamping system should exert only the pressure necessary to stabilize the biomodel.

10.2. Anatomical orientation of the biomodel

Correct anatomical orientation of the model is essential for simulation activities to consistently replicate the typical perspective of the surgeon or student while manipulating the heart.

When installing the biomodel on the base, it is recommended to verify the following anatomical landmarks:

- The cardiac apex should be oriented toward the lower region of the system.
- The great vessels (aorta and pulmonary artery) should be oriented toward the upper region of the model.
- The anterior surface of the heart should face the user's access area.

This orientation allows the model to reproduce an anatomical position approximating that of the heart within the thoracic cavity, facilitating the spatial interpretation of cardiac structures during demonstrations or educational exercises.

When using models with complex congenital heart defects, it may be advisable to slightly adjust the orientation to improve visibility of the regions of interest.

10.3. Installation of the Thoracic Cover

Once the biomodel is installed on the support base, the structural thoracic cover can be attached; this cover provides a simplified representation of the pediatric thoracic cavity and offers greater anatomical context during simulation.

To install it, align the cover's attachment points with the structural base and press gently until it is securely fastened. Once installed, the cover should be firmly in place without interfering with access to the biomodel.

This component partially limits the user's field of view, helping to create handling conditions that more closely resemble a real surgical environment.

10.4. Final System Verification and Adjustments

After completing the system assembly, it is recommended to perform a brief verification to ensure the proper functioning of the assembly.

The following aspects should be checked:

- The biomodel remains firmly secured within the clamping mechanism
- There is no excessive compression of the elastomeric material
- The model maintains the correct anatomical orientation
- The thoracic cover is properly attached
- There are no loose or unstable components
- The user can access the biomodel without structural interference

It is also recommended to perform a light manipulation test, simulating typical movements during educational or demonstration activities, to confirm that the system maintains its stability.

Once these aspects have been verified, the system is ready for use in simulation activities, anatomical teaching, or clinical demonstrations within educational or research contexts.

11.Final notes and recommendations

This guide describes a reproducible workflow for the fabrication and integration of anatomical cardiac biomodels using additive manufacturing. Proper handling and storage of the system components help preserve the quality of the biomodel and facilitate its use in educational or demonstration activities.

11.1. Storage of Biomodels

Due to the elastomeric nature of Silicone 40A, the material may exhibit slight surface stickiness, which facilitates the accumulation of dust or particles during storage. To preserve the biomodel prior to use, it is recommended to:

- Store the biomodels in closed containers or protective cases.
- Keep them away from dust or lint.
- Avoid prolonged exposure to direct sunlight, as UV radiation can accelerate the degradation of the material.

When planning educational activities that involve handling, cutting, or suturing, it is recommended to have several pre-made biomodels available, as these types of procedures permanently alter the model's geometry.

11.2. Use of the biomodel

Biomodels can be used in different educational contexts depending on the type of activity. In anatomical demonstrations or educational exhibits, the model can be handled without altering its structural integrity. However, when the biomodel is used for cutting or suturing practices, the model should be considered single-use, as these procedures permanently alter the elastomeric material.

11.3. Final Consideration

The availability of clear technical guidelines for the fabrication of anatomical biomodels helps facilitate the incorporation of physical simulation tools into educational settings. Using the workflow described in this document, it is possible to transform digital models into physical representations that support three-dimensional understanding of cardiac anatomy and enrich teaching and demonstration activities in the field of medical education.